

Relationships of Lower Body Maximum Strength, Power, Speed Field Tests with Performance of Men Javelin Throwers

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ABSTRACT

This study aimed to investigate relationship of performance of men javelin throwers with lower body maximum strength, as measured by 1RM barbell back squat, peak leg power assessed by vertical jump using Noval (William et al, 2012) equation and acceleration and speed testing with 30m sprint dash. Thirty national and international level javelin throwers were examined for this study. Tests were conducted during competition period only. The results indicate that javelin throw performance is highly related to 1RM barbell squat maximum strength test (0.768, $p < 0.01$) and peak leg power (0.573, $p < 0.01$). Javelin throw performance has not shown any significant relationship with 30m sprint test for acceleration and speed. Further javelin throwers were divided into two groups above and below 70m of performance levels. Group with performance level below 70m has shown significant relationship with 1 RM barbell squat maximum strength test (0.639, $p < 0.01$) only; whereas performance of Group above 70m is strongly related to peak leg power (0.676, $p < 0.01$). Results suggest that maximum strength contributes to the power out put and performance of javelin throw. But its contribution has diminishing relationship with increasing performance level of javelin throw.

Key words

1RM barbell squat, Peak leg power, maximum strength, power

INTRODUCTION

Release velocity is the most important factor in javelin throw performance. A javelin thrower generates kinetic energy in the approach run and transfers energy from the lower extremities to the upper extremities and ultimately to the implement in the final delivery stride. According to Coleman (1998) power for the overhead thrower is primarily generated with leg

extension, hip rotation and trunk flexion. these components generate more than 50% of the forces in a standing throw. Coaches and throwers know the importance of lower body maximum strength and power for better performance in javelin throw. But, very few studies have been conducted in examining the relationship between lower body field tests and performance of javelin throw. Bouhel et al (2007) in a study of male

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javelin throwers found that maximal anaerobic power of both legs and arms were significantly correlated with javelin performance, with legs having higher correlation as compared to arms.

The duration of the javelin pull from start to finish has been measured at 0.16-0.18s (Zatsiorsky, 1995). This indicates that coaches must consider not only velocity of movement, but also total force production and rate of force production. In other words, the athlete should train to produce the greatest amount of force in the shortest amount of time (Young, 2001). In simple terms strength can be defined as the ability to apply force (Siff, 2000). Power can be defined as strength x speed or force x velocity. In both instances, power is measured in watts (W). Muscular peak power (PP) has been designated as the maximum potential product of strength and speed and is demonstrated as the highest power output attainable during a given movement/ repetition (Komi, 1992). As power is the product of strength and speed, maximum strength is the basic quality affecting power output. Potentially, maximum strength could affect peak power because a given load would represent a smaller percentage of maximum; thus, making the load easier to accelerate (Schmidtbleicher, 1992).

Although, it's generally believed that maximum strength should have its greatest effect on power produced at heavy loads, some evidence indicates that maximum strength influences power over a much wider range than might be expected (Stone et al 2003).

In accordance with the concept of above cited studies, research is needed to

understand the relationships of field tests of maximum strength, power and speed with performance of javelin throw.

METHODOLOGY

Thirty national and international level javelin throwers were tested for lower body maximum strength as measured by 1RM barbell back squat, peak leg power assessed by vertical jump using Noval equation (William et al 2012): $PP(W) = 63.6 \times VJH$ (centimeters) + $42.6 \times BM$ (kilograms) - 1846.5. Acceleration and speed were tested with 30m sprint dash. Performance was tested with 800gm javelin using complete approach run and standing throw. The mean and standard deviation values along with the correlation coefficient between the variables were computed.

RESULTS & DISCUSSION

Descriptive statistics can be seen in Table 1. Data are offered for performance in javelin, standing throw with javelin, 1RM barbell squat, peak leg power 30m sprint and body weight with mean and standard deviation.

Table 2 shows relationship of selected variables of lower body maximum strength, power and speed with performance of javelin throw. Javelin throw performance has shown strong correlation with 1RM barbell back squat ($r = 0.768$), Peak leg power ($r = 0.573$), and standing throw javelin ($r = 0.781$) at a value of $p < 0.01$. Javelin throw performance has not shown significant correlation with 30m sprint. 1RM barbell back squat has shown significant correlation with peak leg power ($r = 0.679$) and standing throw ($r = 0.755$). 30m sprint has also not shown significant correlation with peak leg power and

standing javelin throw. Peak leg power has shown significant correlation with standing throw javelin ($r = 0.607$) at a value of $p < 0.01$.

Table 3 shows the correlation of javelin throw performance with selected variables of Group I (above 70m performance) and Group II (below 70m performance). Javelin throw performance of Group I shows significant relationship

with 1 RM barbell back squat ($r = 0.548$) at value of $p < 0.05$ and peak leg power ($r = 0.676$) at value of $p < 0.01$. Javelin throw performance does not show significant relationship with 30m sprint. Javelin throw performance of Group II shows significant relationship with 1RM barbell back squat ($r = 0.639$) at value of $p < 0.01$; but, shows no significant relationship with 30m sprint and peak leg power.

Table –1 : Mean and SD values of selected variables

S. No.	Tests	Mean	SD
1.	Performance Javelin (m)	68.85	7.30
2.	Standing throw javelin (m)	49.55	7.87
3.	1RM Barbell back squat (Kg)	150.83	38.70
4.	Peak Leg Power (N)	5775.14	1086.09
5.	30m Sprint (s)	3.92	0.29

Table –2 : Mean and SD values of selected variables

	Javelin performance	1 RM barbell back squat	30m sprint	Peak leg power
1 RM barbell back squat	0.768**			
30m sprint	0.036	0.044		
Peak leg power	0.573**	0.679**	-0.165	
Standing throw javelin	0.781**	0.755**	-0.017	0.607**

*Significant at $p < 0.01$ (Two tailed)

Table - 3: Relationship of javelin throw performance with selected variables for Group – I and Group – II

	Above 70m Group- I (N = 14)	Below 70m Group- II (N = 16)
1 RM barbell back squat	0.548*	0.639**
30m sprint	0.101	0.037
Peak leg power	0.676**	0.368

*significant at $p < 0.05$, **Significant at $p < 0.01$ (Two tailed)

The findings of this study indicated that there is a positive interaction between javelin throw performance and lower body maximum strength and power field tests. Current data support this conjecture, as demonstrated by very high significant correlation of javelin throw performance with 1RM barbell back squat ($r = 0.768$) and peak leg power ($r = 0.573$), at the value of $p < 0.01$. 1 RM barbell back squat has shown significant relationship with peak leg power ($r = 0.679$) at the value of $p < 0.01$.

Conceptually, the results agree with the views of Young(2001) that there is greater need to develop leg strength for javelin event. Strength development will enable the athlete to better overcome these loads so that they can proceed into the throwing motion with more velocity. According to Zatsiorsky (1995) increased speed developed in the approach of the javelin throw places a considerably greater load on the athletes' legs (when they hit the power position).

30m sprint doesn't show significant relationship with javelin throw performance and other parameters. Sprint ability over short distances (under 10m) and longer distances (over 30m) are considered by many researchers and practitioners to require separate and specific strength qualities (Delecluse et al, 1995; Young, Benton, Duthei & Pryor, 2001). It has been proposed that concentric strength measure should better relate to starting ability (up to 10m) due to this phase having a greater concentric component as compared to later in the sprint where SSC activity is more important (Young et al, 2001). It is also generally considered that shorter sprints require greater contributions of knee

extensor activity versus longer sprints that are characterized by greater hip extensor activity (Komi, 1992).

It is also of interest to note that Group I (above 70m performance) has shown significant correlation with 1RM barbell back squat and Peak leg power. On the contrary Group II (below 70m performance) has shown significant correlation with only 1RM barbell back squat. As Cronin et al (2005) and Stone et al (2005) stated that most of the research studies those found significant correlations between maximum strength and power has been conducted with novice subjects or lesser trained subjects. Research examining more experienced subjects has found conflicting results and poses a potentially diminishing association between the magnitude of dynamic force production and power related performance measures. The results of this study also indicate that as the level of performance increases the association between lower body power and performance increases, but relationship between maximum strength and performance diminishes.

CONCLUSION

The purpose of the study was to examine relationships of field tests of maximum strength, power and speed with performance of javelin throw. The significant relationship of javelin throw with lower body strength and power variables provide the evidence that practice of focusing on maximum strength and power by coaches and expert is true. As the performance increases relationships of javelin throw performance with maximum strength diminishes. There is significant relationship between lower body power and

javelin performance at higher level of javelin throw performance.

PRACTICAL APPLICATIONS

From the present study it appears that lower body maximum strength plays a decisive role for the development of power

and throwing performance. But, as the level of performance increases, training should be more focused on power development. Effect of sprinting on performance still needs more investigation. Throwers can be tested for shorter distances.

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